Isolating Emotion from Motion

Classifying Affective States from Wearable Physiological Signals, and Testing Whether Body Motion Confounds the Model

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1. Motivation

Affective computing seeks to let machines recognize human emotion. Wearables make this cheap and continuous , but only if it works away from the laboratory.

- Emotion-recognition systems infer affective state from physiological signals such as cardiac activity (ECG/PPG) and electrodermal activity (EDA).
- On consumer wearables these signals are captured by optical PPG sensors, which are highly susceptible to motion artifacts from ordinary movement.
- Most published models are trained and evaluated on clean, stationary lab recordings, casting doubt on their real-world validity.

• Central risk:
a model may associate movement itself with an emotion rather than the underlying physiology, a shortcut that would fail in deployment.

• Beneficiaries:
clinicians and patients in mental-health care, and everyday users of affordable wrist-worn stress trackers.

2. Research Objectives

• Research question —
can physiological signals classify emotional state under realistic, subject-independent conditions, and to what extent does body motion contribute to that prediction?

• Objective 1 —
build and rigorously validate a three-class classifier (baseline, stress, amusement) on the WESAD dataset.

• Objective 2 —
quantify the role of motion through feature ablation and model-derived importance.

• Expected outcome —
a validated classifier and direct evidence on whether motion is a genuine predictor or a confound.

3. Background & Related Work

- A systematic review of 92 studies found ECG and PPG the dominant modalities for emotion recognition, with SVM the most common classifier and reported accuracies frequently above 90% (Ismail et al., 2024).
- The WESAD benchmark reported ≈ 80% accuracy for three-class affect and ≈ 93% for binary stress detection using hand-crafted features (Schmidt et al., 2018).
- Wearable PPG is known to degrade under movement; sensor-accuracy studies attribute substantial error to motion artifacts (Bent et al., 2020).

• Research gap:
reported performance overwhelmingly comes from controlled, low-motion settings, and few studies test directly whether motion drives the prediction. This project isolates that question.

4. Dataset

• Source —
WESAD (Schmidt et al., ICMI 2018): 15 participants; RespiBAN chest unit and Empatica E4 wrist unit.

• Conditions —
neutral baseline, stress (public speaking + mental arithmetic), and amusement (funny video clips).

• Processing —
chest signals at 700 Hz segmented into non-overlapping 60-second windows; each reduced to 14 statistical features.

• Resulting data —-
520 windows; baseline 283, stress 155, amusement 82; numerical inputs, and there were no missing values.

5. Exploratory Data Analysis

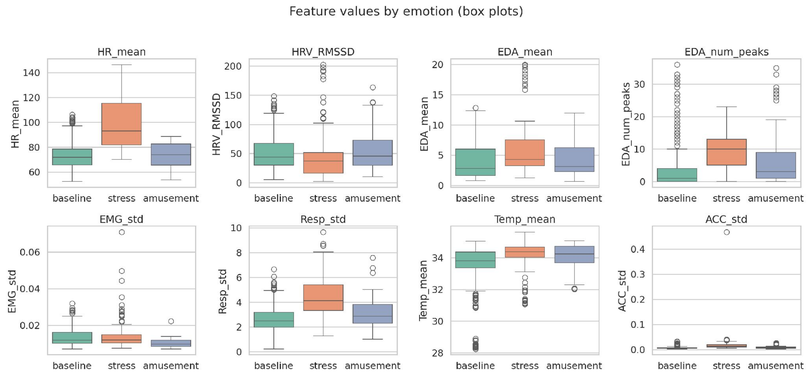


Fig 1. Feature values by emotion. Stress is elevated across heart rate, respiration variability, and EDA; amusement overlaps baseline.

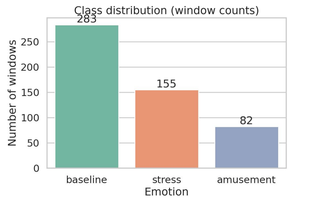


Fig 2. Class imbalance, 3.4 : 1.

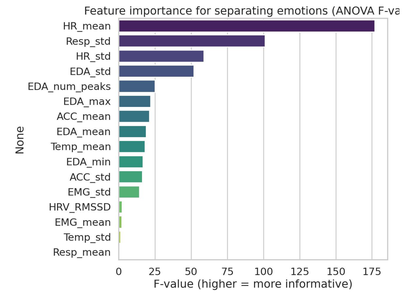


Fig 3. ANOVA ranking: motion (ACC) 7th of 16.

- **Stress is physiologically distinct; amusement overlaps baseline, anticipating classification difficulty.**
- **Univariate ANOVA ranks heart rate first and places motion only 7th of 16, an early sign motion is not the dominant cue.**
- **Redundancy:**

EDA mean/min/max correlate at r ≈ 1.00; ACC_std at r = 0.94 with temperature variability.

6. Machine-Learning Methodology

• Preparation —
dropped two redundant EDA features (r = 1.00); standardized all features to zero mean and unit variance.

• Validation — Leave-One-Subject-Out:
train on 14 participants, test on the held-out participant, rotating through all 15. This prevents identity leakage from inter-individual differences and estimates generalization to unseen users.

• Metric — macro-F1
(unweighted mean over classes) as primary, since accuracy is inflated by the majority class under imbalance.



Fig 4. Analysis pipeline, from raw signal to subject-independent evaluation.

• Logistic Regression
— interpretable linear baseline.

• Random Forest
— non-linear; yields feature importance.

• SVM (RBF)
— standard in the ERS literature; effective on modest samples. All models use balanced class weights.

7. Results

Table 1. Model performance (LOSO)

Model	Acc.	F1
Logistic Reg.	0.648	**0.623**
Random Forest	**0.731**	0.577
SVM (RBF)	0.629	0.616

Table 2. Per-class (Logistic Reg.)

Class	P	R	F1
baseline	0.79	0.58	0.67
stress	0.71	0.79	0.74
amusement	**0.37**	**0.61**	**0.46**

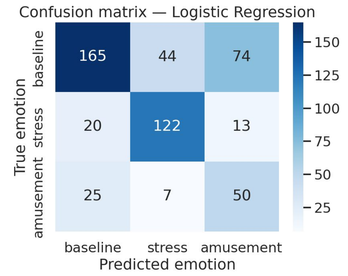


Fig 5. Confusion matrix, Logistic Regression.

• Accuracy misleads.
Random Forest tops accuracy (0.73) yet has the lowest macro-F1 (0.58): it nearly abandons amusement (recall 0.07) and follows the majority class. Macro-F1 exposes this which means it is the appropriate metric.

• Best balanced model:
Logistic Regression (F1 0.623) the simplest model is also the fairest.

• Amusement is hardest.
74 baseline windows are misread as amusement (Fig 5), matching their physiological overlap.

Testing the motion confound

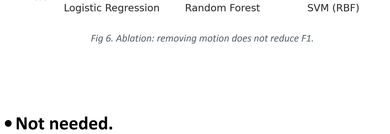


Fig 6. Ablation: removing motion does not reduce F1.

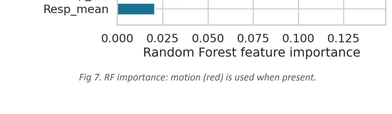


Fig 7. RF importance: motion (red) is used when present.

• Not needed.
Removing the motion features leaves macro-F1 unchanged (marginally higher) for every model (Fig 6) — the classifier does not depend on motion.

• Yet used.
When present, Random Forest ranks motion (ACC_std) its 2nd most-used feature (Fig 7).

• Reconciliation , motion is used but not needed.
ACC_std correlates r = 0.94 with temperature variability, so removing it lets correlated physiological features carry the same signal with no loss.

• Takeaway.
The feared motion shortcut does not compromise performance on WESAD: the model classifies from genuine physiology and treats motion as a redundant cue rather than a crutch. High statistical importance ≠ necessity.

8. Conclusions

- Physiological signals classified three affective states at macro-F1 0.62 under strict subject-independent validation, a conservative, realistic estimate.
- Stress detection is robust; amusement remains difficult due to physiological similarity to rest.

• The classifier does not depend on body motion
performance is preserved when motion is removed. Although it does use motion, redundantly, when present.

9. Limitations & Future Work

• WESAD is largely stationary
the ablation result may not extend to high-movement activity.

- Only 520 windows from 15 subjects; amusement has 82 examples.
- 60 s windows may be too short for reliable HRV; hand-crafted features only.

• Next:
re-test on a high-motion set (PPG-DaLiA); add gradient boosting / neural models for amusement; frequency-domain features; on-device deployment.

10. References

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Tools: Python, pandas, NumPy, Matplotlib, seaborn, SciPy, NeuroKit2, scikit-learn, Google Colab.